

Paper story: experiments

July 23, 2026

This is a progress report: the story of the paper told through the figures we have produced so far. It is organised by PDE — steady-state Poisson first, then the time-dependent Allen–Cahn stepper — and closes with what comes next (Stokes). **I am sorry for the language, but making Claude make this is so much faster.**

1 Poisson

1.1 Least-squares loss and spectral preconditioning

- Physics-informed neural operators often **do not train**, and the culprit is *spectral ill-conditioning* inherited from the unbounded PDE operator: the loss landscape is a long, narrow valley that first-order optimisers cannot descend.
- To demonstrate this cleanly we use the **least-squares residual loss** based on the stiffness matrix A on a $Q1$ finite element space. We insert a parametrised preconditioner P_t ,

$$\mathcal{L}(\theta) = \frac{1}{2} \|P_t (A u_\theta - b)\|_2^2, \quad P_t = (1 - t)I + tA^{-1},$$

so $P_0 = I$ (no preconditioning) and $P_1 = A^{-1}$ (full preconditioning), with a convex interpolation in between. The interpolation is made tractable either through the discrete sine transform (exact on the regular grid) or a dense eigenvalue solve.

- The sweep over t (Figure 1) shows the two extremes: **without** preconditioning ($t = 0$) training **stalls**; with **full** preconditioning ($t = 1$) we **recover the gold standard of data-driven training**. The reason is the squaring: the least-squares Hessian conditioning scales like h^{-4} — already $\approx 6.5 \times 10^5$ on a 64^2 grid, which is horrible.
- In practice A^{-1} is never formed; a **multigrid** V-cycle supplies $P \approx A^{-1}$ at *optimal complexity*. The multigrid-preconditioned run matches data-driven training (rightmost curve in Figure 1). The whole story is summarised by plotting the final validation error *against the condition number* (Figure 2): the runs collapse onto a single monotone curve, error rising with conditioning.

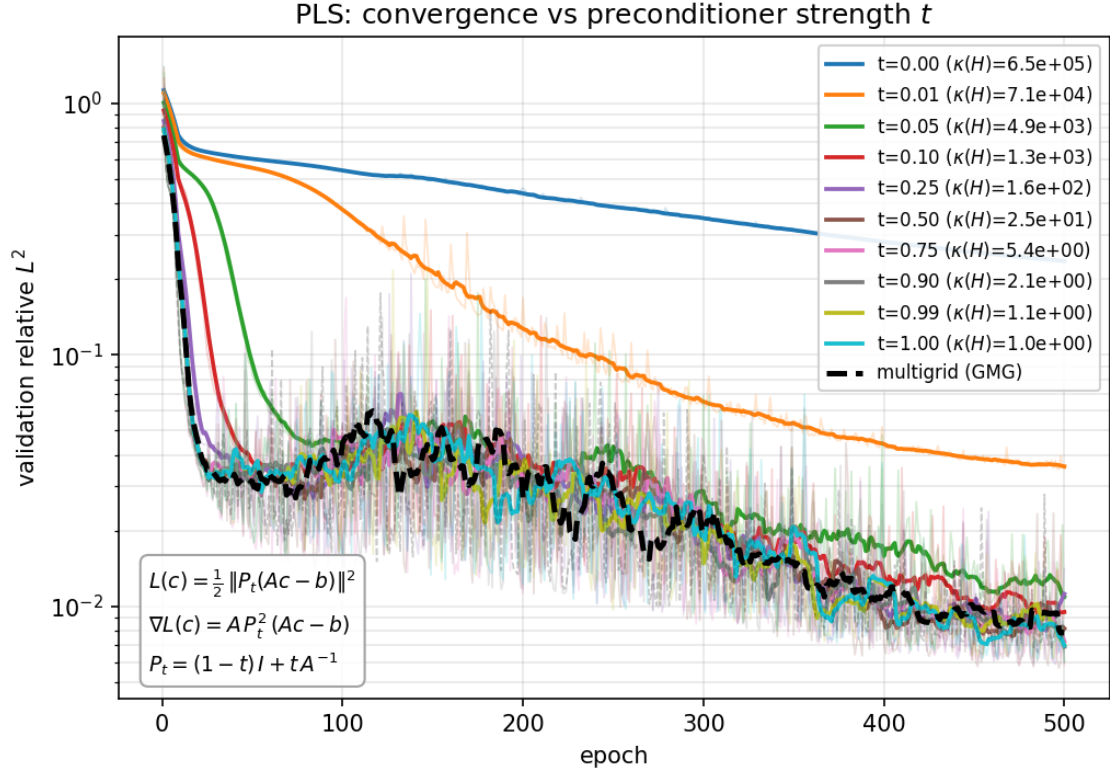


Figure 1: Poisson, least-squares loss. Validation error vs. epoch as the preconditioner strength t is swept from 0 (identity) to 1 (A^{-1}), plus the multigrid V-cycle and the data-driven reference. Unpreconditioned training stalls; full / multigrid preconditioning recovers the data-driven gold standard.

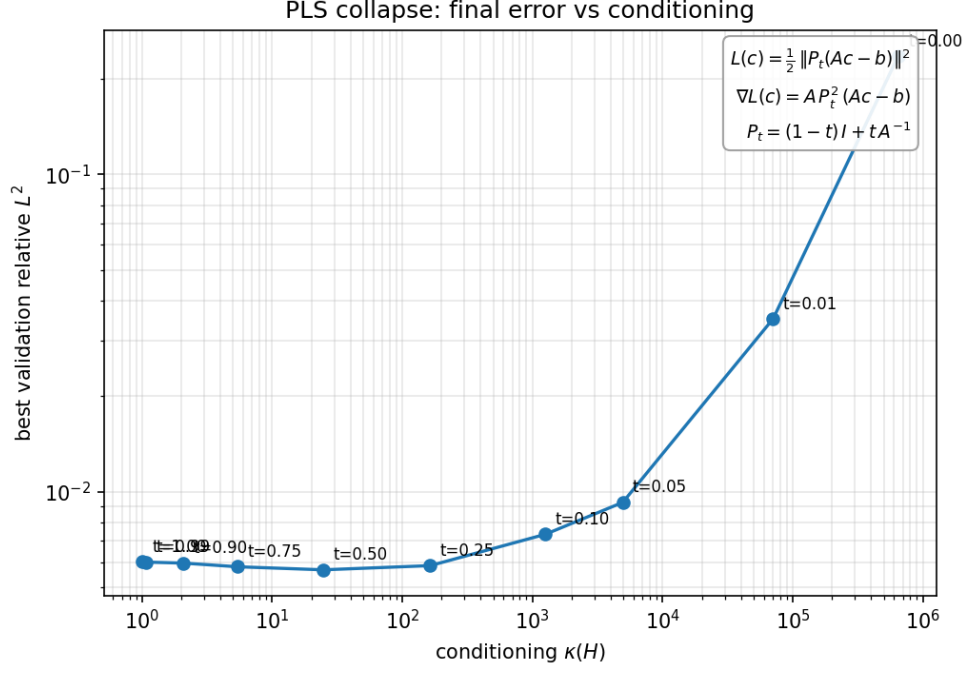


Figure 2: The same runs, final validation error against the condition number of the preconditioned operator. Error grows monotonically with conditioning — the mechanism, made explicit.

1.2 Deep Ritz: the variational formulation is milder

The story changes for the **variational (Deep Ritz) formulation**: its conditioning scales like h^{-2} rather than h^{-4} . This is mild enough that on a 64^2 grid training **succeeds without any preconditioning** (Figure 3). Preconditioning can still be added and yields a small further speed-up — it reduces the conditioning from ≈ 800 down to 1. (On finer grids the h^{-2} growth does become a real problem; there is a figure for that, but it is not important enough to include here.)

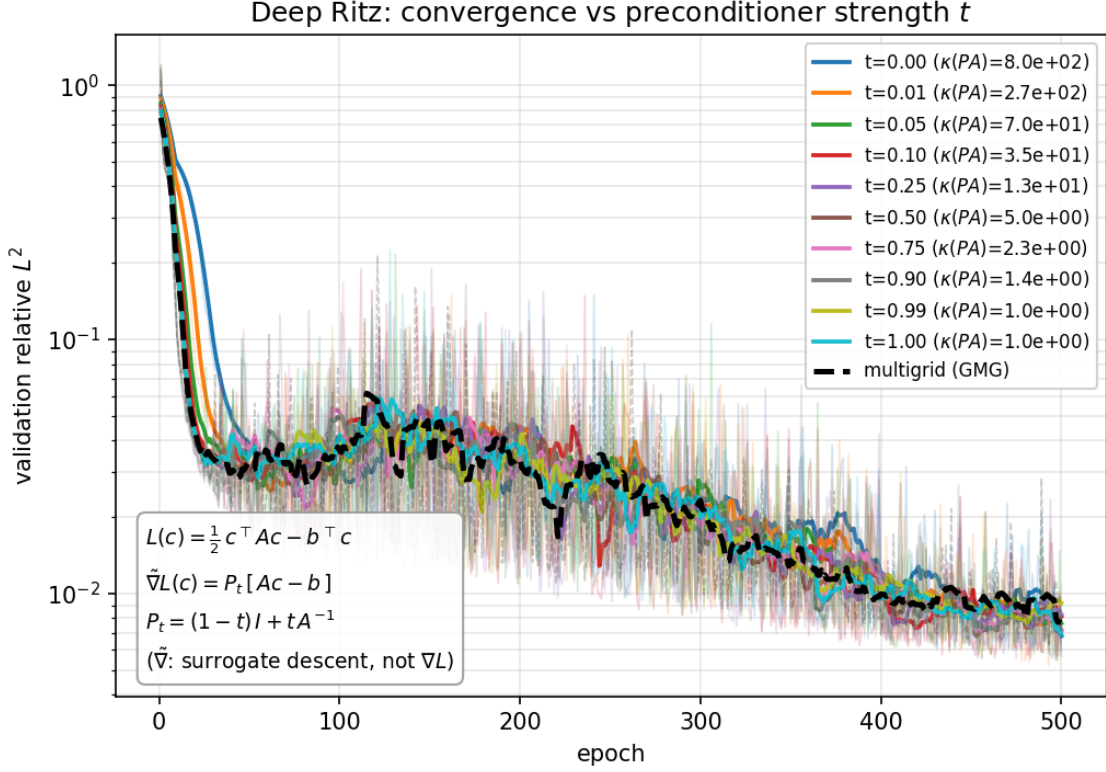


Figure 3: Poisson, Deep Ritz (variational) loss. Because the conditioning is only h^{-2} , training already succeeds unpreconditioned on 64^2 ; preconditioning gives a modest speed-up.

1.3 The infinite-data limit

Because we generate **no labelled data**, we are not bound to a fixed dataset. For Figure 4 we fix the optimisation budget (500 epochs’ worth of forward passes at a dataset of 1024) and grow the training-set size from 32 up to 4096. Crucially, we also add the **infinite dataset** — one that never revisits a sample — as the dashed black line.

This is a genuine selling point of physics-informed operator learning: you want to maximise what you extract from your samples, and *never revisiting a sample is an option*. We believe one can show theoretically that this **eliminates one of the classical errors** in the statistical- learning error decomposition — the estimation error tied to the finiteness of the dataset.

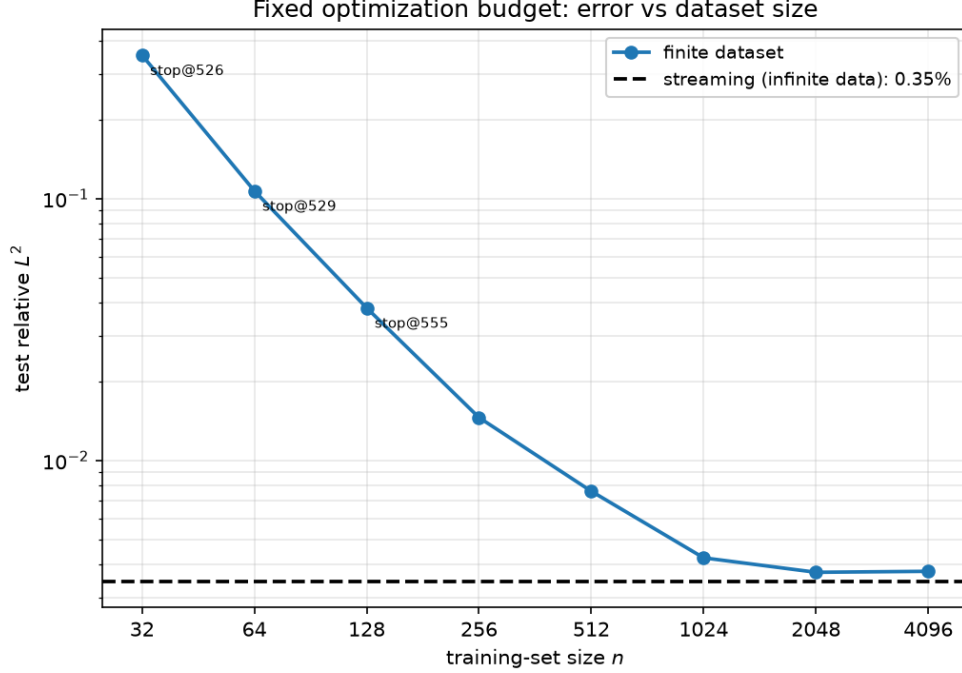


Figure 4: Test relative L^2 vs. training-set size at a fixed optimisation budget. Error decreases with dataset size and saturates at the infinite-data (streaming, never-revisit) limit, the dashed black line.

2 Allen–Cahn

We now move to the time-dependent setting: an FNO trained as a one-step Allen–Cahn integrator and rolled out autoregressively.

Opening question — backprop through time. Do we know the best way to handle backpropagation through the unrolled rollout? Here we **detached the inputs and never backpropagated through the unrolled operator** — the *pushforward trick* (Brandstetter et al.). It performed best across all losses in our tests, and it is cheaper. This deserves discussion, but it is our current default.

Setup. We train on a **10-step** rollout, 128^2 grid, $dt = 0.01$, and reaction strength $\varepsilon = 32$ — note ε enters the equation *squared*, and here it is large, not small. Data generation now defaults to a **convex–concave (Eyre) splitting** on the 128^2 grid at this time step, and we train the FNO in exactly the same setting. From that splitting we derive two physics-informed losses: the natural **minimizing-movement** objective, and its **least-squares** variant. We abandoned the implicit-Euler scheme. Finally we benchmark against **data-driven** training. The least-squares variant is preconditioned with multigrid again — the problem is nonlinear, but we hand-wave the Jacobian into a constant operator so multigrid applies.

Findings.

- Physics-informed **least squares without preconditioning fails** (those curves are not in the figure yet).
- The **other three all work quite well** (Figure 5). Minimizing movement appears to have an edge, though that may be down to learning-rate details.
- **Rollouts stay stable out to 100 steps** even though the models were trained on only 10 steps — which is quite nice.

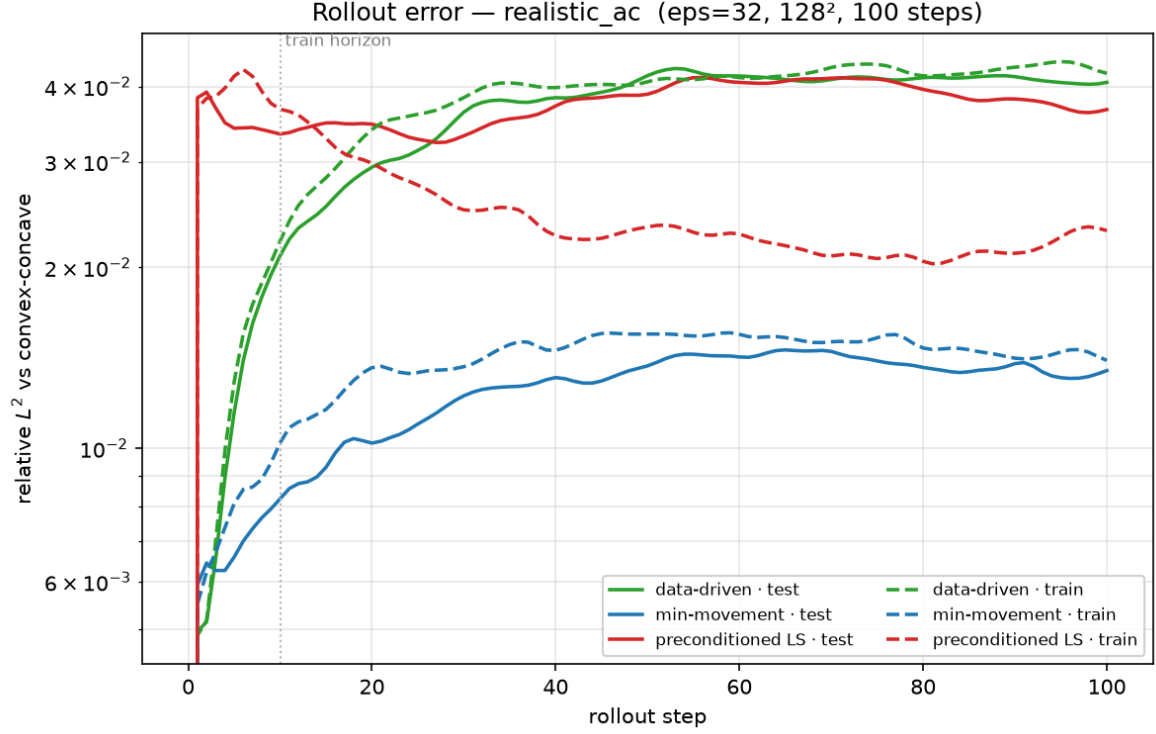


Figure 5: Allen–Cahn, relative L^2 vs. the convex–concave reference along the rollout (solid = test, dashed = train), one colour per training loss. Trained on 10 steps, rolled out to 100; the error stays controlled well past the training horizon.

For a qualitative feel, Figure 6 shows the 2D fields: rows are rollout timesteps, columns are the reference next to each learned stepper.

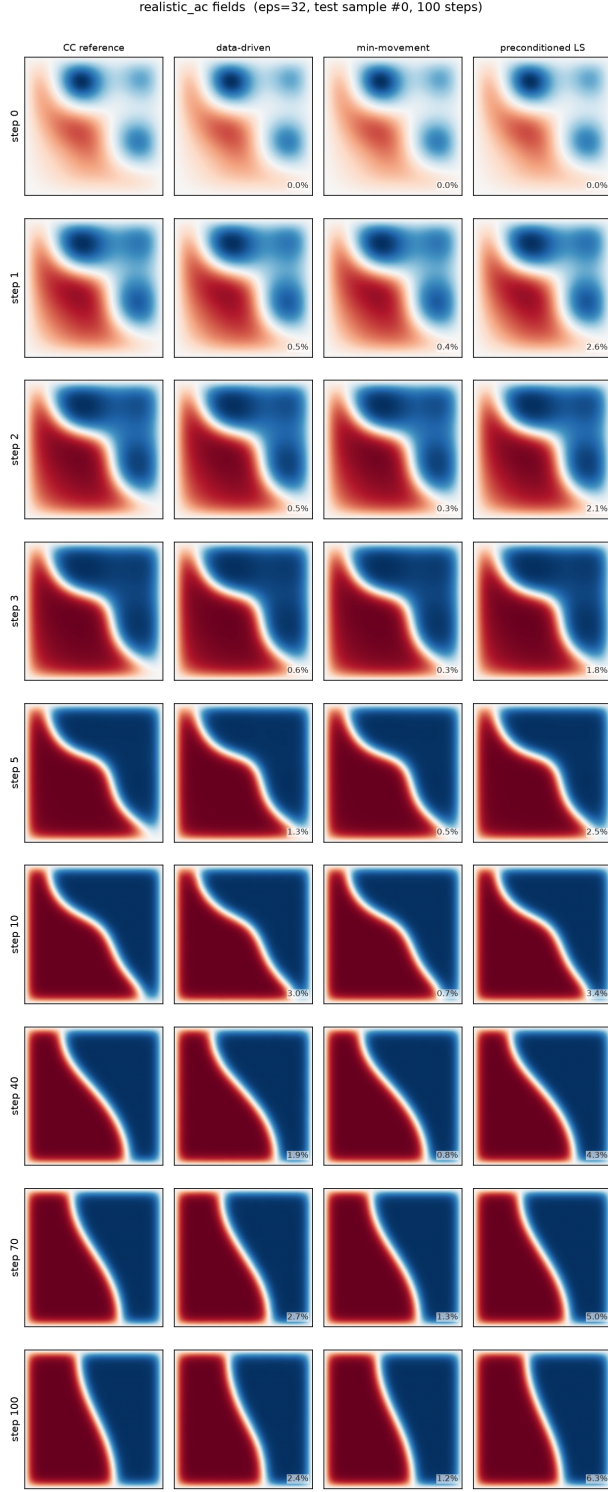


Figure 6: Allen–Cahn rollout fields for one test sample: rows are timesteps, columns are the convex–concave reference and each trained model, annotated with the per-panel relative L^2 .

3 Future plans

Stokes. First results using a **block-diagonal preconditioner** are promising. We expect this to land next week.

Fine-tune Poseidon We can finetune Poseidon physics informed for some PDEs, certainly the Poisson.